

SOC Estimation for Lithium-Ion Battery Cell Using Extended Kalman Filter

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Abstract— The State of Charge (SOC) estimation is an essential part of a Battery Management System. Nevertheless, because the discharge and charging of battery cells requires complicated chemical operations, therefore, it is hard to determine the state of charge of the battery cell. In this paper, a lithium iron phosphate battery cell with 8 Ah capacity and 3.2 Volt rated voltage was studied. The battery is modelled to reflect the dynamic of the battery encompassing mainly four elements; an Open Circuit Voltage (OCV) source, two RC network, and one resistor. The model parameters are identified by using Forgetting Factor Recursive Least Mean Squares and time domain extraction method. Parameters are converged to their real values and these values are used to estimate the state of charge of the cell using Extended Kalman Filter algorithm based on battery model dynamic. The results in our study show that SoC which is estimated by the implemented Extended Kalman Filter converges to battery's real SoC value. It is also shown that the Extended Kalman Filter update and prediction stages iterations move forward to minimize the error between real SoC and estimated SoC.

Keywords— Lithium battery modelling; Lithium battery parameter identification; Extended Kalman Filter (EKF); state-of-charge (SoC);

1. Introduction

The use of lithium batteries is increasing year by year. Hazardous gas emitting vehicles require the usage of battery. Most countries endeavor and fund money to achieve zero emission of fuel-consuming vehicles. There are many battery types produced until now and prototyped. Among them, lithium cells are substantially used in the fields of BESS (Battery Energy Storage System), HEV (Hybrid Electric Vehicles), EV (Electric Vehicle), portable hardware such as PCs, mobile phones, tablets and so forth. [1]. In brief, lithium batteries are all over the place. Lithium iron phosphate batteries are chosen because of their long service life, excellent reliable quality and cheap price. [2].

The rest of the paper is organized as follows. Section II introduces the experimental setup of the work. Section III covers lithium iron phosphate battery modelling and parameter identification. SoC estimation and EKF based SoC estimation are studied and proposed in Section IV and V respectively. Experimental results are discussed in Section VI. The

outcomes are summarized as theoretical and experimental results in Section VI.

2. Experimental Setup

The test bench setup is shown in Fig. 1. It consists of a battery (HW-38120HP), an electronic load to draw constant current from the battery cell, a computer for the human-machine interface, experimental data acquisition system and the battery cell. The battery testing system is responsible for discharging the battery cell and record the current and voltage data. The host computer is used to obtain voltage and current data from the DAQ 34970A (Data Acquisition System) and analyze it. The measurement tolerance of the measurement sensor inside N3300A device is within 0.25%. The temperature of the test room is set to 25 degrees.

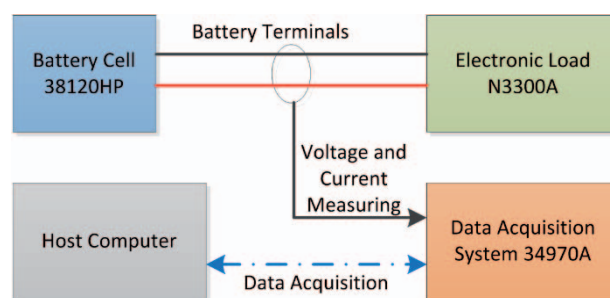


Fig. 1 Experimental setup

3. Battery Model and Parameter Identification

3.1 Battery Model

There are several equivalent circuit models for the lithium battery cell such as Shepherd model, Unnewehr universal model, Nernst model, dual polarization model, zero state hysteresis model and enhanced self-correcting model [3]. The Dual Polarization (Second Order) model is selected as the equivalent model for the battery cell. Dual Polarization battery cell model is shown in Fig. 2. Using the resistors and capacitors in the model, the aging status of the battery can be represented with this equivalent circuit model.

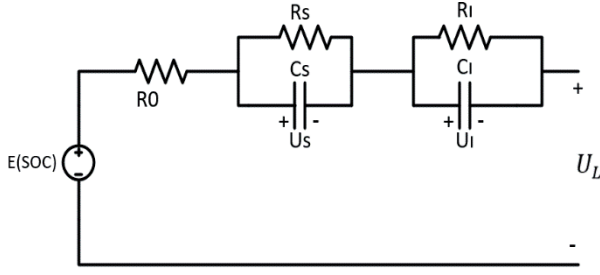


Fig. 2 Dual Polarization (2 RC) Thevenin Model

RC branches of the above model can be increased. However, this complicates the dynamic of the model and does not provide additional benefit. Therefore, Two-RC model is optimal and quite suitable for modelling lithium battery cell behavior.

The lithium battery is a power source. In the above model, controlled voltage source E_{SOC} is connected in series with the internal resistance R_0 . In order to reflect the polarization effect of the lithium cell, two parallel RC circuits are also connected in series to the controlled voltage source as can be seen in Fig 2. R_s and R_l are polarization resistors, C_s and C_l are polarization capacitors in the model. E_{SOC} is the dependent voltage source of the battery cell, in other words, it is a function of SoC. i is the current flowing through the battery and U_L is the battery terminal voltage which is measured at both ends of the battery. One of the parallel RC branch represents short relaxation time constant (R_s and C_s) and other parallel RC branch represents for long term (R_l and C_l) relaxation time constants [3].

The dynamic equations of the model can be derived by using Kirchhoff's Law as in (1) and (2):

$$\dot{u}_l = \frac{1}{C_l} i - \frac{1}{R_l C_l} u_l \quad (1)$$

$$\dot{u}_s = \frac{1}{C_s} i - \frac{1}{R_s C_s} u_s \quad (2)$$

Using (1) and (2), State Space representation of two RC equivalent model can be easily obtained as below. Here, the state variable representing *SoC* is defined as s_{oc} :

$$\dot{x}(t) = Ax(t) + Bu(t) \quad (3)$$

$$y = Cx(t) + Du(t) \quad (4)$$

Where;

$$x(t) = \begin{bmatrix} s_{oc} \\ u_l \\ u_s \end{bmatrix}; \quad A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -\frac{1}{\tau_l} & 0 \\ 0 & 0 & -\frac{1}{\tau_s} \end{bmatrix}; \quad B = \begin{bmatrix} -\frac{1}{C} \\ -\frac{1}{C_l} \\ -\frac{1}{C_s} \end{bmatrix} \quad (5)$$

$$\tau_l = R_l C_l \quad (6)$$

$$\tau_s = R_s C_s \quad (7)$$

$$D = R_0; \quad C = \begin{bmatrix} OCV \\ s_{oc} & -1 & -1 \end{bmatrix} \quad (8)$$

In accordance with the two RC model shown in Fig 2. The output voltage of the circuit can be calculated as:

$$U_L = E_{SOC} - i R_0 - u_s - u_l \quad (9)$$

Taking noise into account sensor and process noise, it is possible to obtain a discretized system model as defined below [3]:

$$x_{k+1} = A_d x_k + H i_k + w_k \quad (10)$$

$$U_{Lk} = E(s_{ock}) - u_{sk} - u_{lk} - i_k R_0 + v_k \quad (11)$$

where ;

$$A_d = \begin{bmatrix} 1 & 0 & 0 \\ 0 & e^{-T/\tau_l} & 0 \\ 0 & 0 & e^{-T/\tau_s} \end{bmatrix} \quad (12)$$

$$B_d = \begin{bmatrix} -T/C \\ \tau_l / C_l (1 - e^{-T/\tau_l}) \\ \tau_s / C_s (1 - e^{-T/\tau_s}) \end{bmatrix} \quad (13)$$

s_{oc} is the battery state of charge, i_k is the current, u_{sk} and u_{lk} are the polarization voltage of the battery, U_{Lk} is the battery cell voltage which is output of the model, T is the sampling interval and C is the nominal capacity of the battery cell.

The process noise and the measurement noise are specified as w_k and v_k respectively. Both of them are independent Gaussian white noise. The mean value of w_k is zero and the covariance matrix of w_k is Q_k . The mean value of v_k is zero and the covariance matrix of v_k is R_k .

3.2 State of Charge (SoC)–Open Circuit Voltage(OCV) Curve

Open circuit voltage of the battery cell is dependent on the state of charge. By applying current pulses to the battery cell, SoC from the full charged cell is reduced by 10%. These pulses have 0.8 Amperes magnitude and applied for 1 hour to the battery cell. Battery terminal voltage under current pulse discharge test can be seen from Fig. 3.

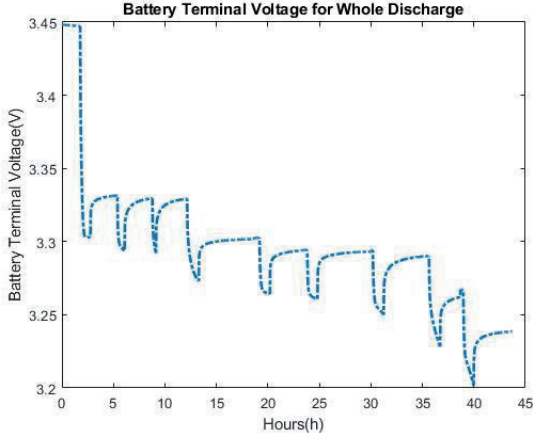


Fig. 3 Battery terminal voltage while discharging

At the end of the resting periods, voltages of the lithium cell are obtained and registered. These voltage values are used to plot SoC-OCV curve in Fig. 4 and fitted by using curve fitting toolbox of MATLAB.

Battery terminal voltage under current pulse discharge test can be seen from Fig 3. Same process is applied to the battery cell to plot charge curve of OCV-SOC. The curve used in the algorithm is the average of charging and discharging curves.

Average SOC-OCV curve is obtained as:

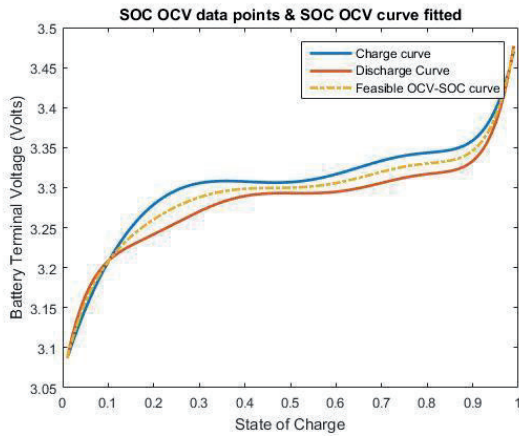


Fig. 4 SOC-OCV curve of the battery cell

The OCV values corresponding to different SoC values are determined and results are shown in Table 1.

Table 1. SOC&OCV Values

SOC	0.99	0.9	0.8	0.7	0.6	0.5	0.4	0.3	0.2	0.1
OCV	3.47	3.33	3.32	3.30	3.29	3.29	3.29	3.26	3.24	3.1

OCV-SOC curve is fitted as a seventh degree polynomial as given in (14). The coefficients in this fitted polynomial is determined through MATLAB simulation and the coefficients are given in (14) as $b(\cdot)$.

$$OCV(s_{oc}) = b(1) + b(2)s_{oc} + b(3)s_{oc}^2 + b(4)s_{oc}^3 + b(5)s_{oc}^4 + b(6)s_{oc}^5 + b(7)s_{oc}^6 + b(8)s_{oc}^7 \quad (14)$$

3.3 Parameter Identification

In this paper, 2 RC equivalent Thevenin circuit is used to estimate SoC of the lithium battery cell. This Thevenin circuit uses the basic circuit components such as resistor, capacitors and so on to simulate the battery cell. If the parameters are calculated correctly, a high accuracy can be achieved in SoC estimation.

In this part of the paper, 1 RC model Thevenin circuit is used to identify parameters of the lithium battery cell. After that, using the parameters found earlier, the other remaining RC parallel circuit parameters are obtained by using time domain extraction method with simple analytic calculations. The reason to do this process is that whole identification of 2 RC parameter brings operational complexity.

Consider linear time-varying SISO system described below in equations (15) -(17) [4]:

$$y(k) = \phi^T(k)\theta + \gamma(k) \quad (15)$$

$$\phi(k) = \begin{bmatrix} -y(k-1), \dots, -y(k-a), u(k-b), \dots \\ u(k-c-d) \end{bmatrix} \quad (16)$$

$$\theta = [a_1, \dots, a_a, b_0, \dots, b_d] \quad (17)$$

In above equations $\gamma(k)$ denotes additive noise at the current sampling point k and a, d and c represents time constants. The aim of the theory is minimizing least square error.

$$V = \sum_{k=1}^L \lambda^{L-k} [y(k) - \phi^T(k)\hat{\theta}]^2 \quad (18)$$

Recursively evaluating $\hat{\theta}$, the minimum value of V can be determined :

$$\hat{\theta}(k) = \hat{\theta}(k-1) + K(k)(y(k) - \phi^T(k)\hat{\theta}(k-1)) \quad (19)$$

$$P(k) = \frac{1}{\lambda} [I - K(k)\phi^T(k)]P(k-1) \quad (20)$$

Usually, forgetting factor λ is set to between 0.9 to 1.

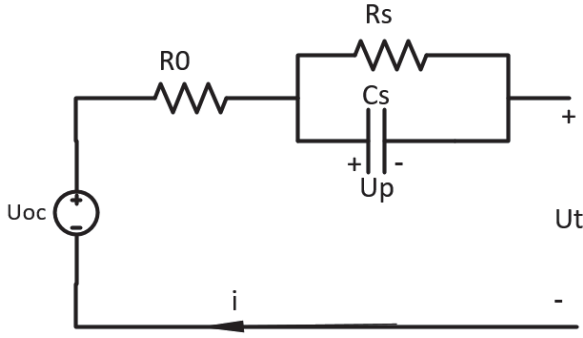


Fig. 5 One RC Thevenin model

It is easy to gather the dynamic of the circuit given in Fig. 5 as in (21) and (22).

$$U_{oc} = U_t + iR_0 + U_p \quad (21)$$

$$C_p \frac{dU_p}{dt} + \frac{U_p}{R_p} = i \quad (22)$$

Laplace transform of the above equations gives us the algebraic equations in (23) and (24):

$$U_{oc}(s) = U_t(s) + I(s)R_0 + U_p(s) \quad (23)$$

$$U_p(s) = \frac{I(s)R_p}{R_p C_p s + 1} \quad (24)$$

Inserting U_p in Eq. (24) into Eq.(23), we have Eq.25.

$$U_{oc}(s) - U_t(s) = I(s)R_0 + \frac{I(s)R_p}{R_p C_p s + 1} \quad (25)$$

Discretizing the equation (25) by applying the method given in [4], we obtain the Equations in (26).

$$\begin{aligned} \frac{R_p C_p}{T} U_t(k) - U_{oc}(k) &= \frac{R_p C_p}{T} U_t(k-1) - \\ \frac{R_p C_p}{T} U_{oc}(k-1) &- \left(\frac{R_0 R_p C_p}{T} + R_0 + R_p \right) I(k) + \\ \frac{R_0 R_p C_p}{T} I(k-1) \end{aligned} \quad (26)$$

Rearranging coefficients in (26), we obtain:

$$\begin{aligned} U_t(k) - U_{oc}(k) &= -k_1 U_t(k-1) + k_1 U_{oc}(k-1) \\ &+ k_2 I(k) + k_3 I(k-1) \end{aligned} \quad (27)$$

FFRLS (Forgetting Factor Recursive Least Squares) technique is utilized to define parameters k_1 , k_2 and k_3 . Then,

R_0 , R_p and C_p can be determined with the help of equations given below.

$$R_0 = -\frac{k_3}{k_1} \quad (28)$$

$$R_p = -\frac{k_2 + R_0}{k_1 + 1} \quad (29)$$

$$C_p = -\frac{\left(\frac{1}{k_1 + 1} - 1 \right) T}{R_p} \quad (30)$$

As shown in Fig. 6, battery cell is resting between B and C points, discharging with 0.8 amperes between A and B points.

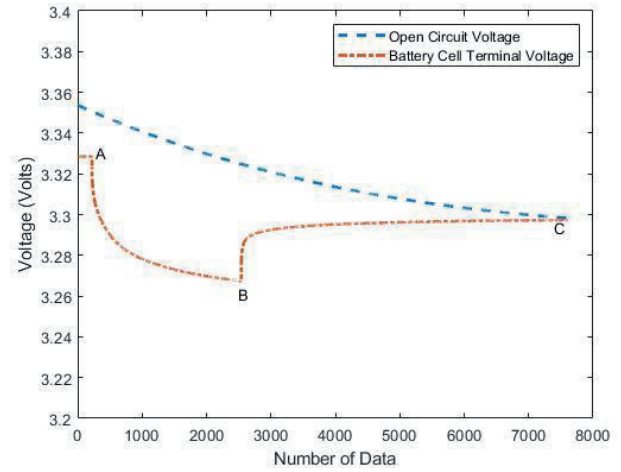


Fig. 6 OCV and battery terminal voltage between certain SoC levels

Thus, voltage of the battery cell and the current drawn is now known. Then, using equation (27) FFRLS algorithm can be implemented to identify parameters in MATLAB platform.

R_0 , R_s and C_s parameters are estimated with this method. Considering the section between A and B, short term time constant is represented by R_s and C_s and R_l and C_l parameters represent the long term time constant.

R_l and C_l parameters are found by using time-domain method separately from R_s and C_s .

For all discharging pulses, short term time constant parameters are calculated. Using the short time constant value, long term time constant parameters are gathered.

$$\tau_s = \frac{\tau_l}{2} \quad (31)$$

This coefficient depends on the test. The long-time constant can be 3 or 4 times higher than the short-time constant, which depends on the tester. A proportional value can also be used while establishing correlation between long and short time constant values.

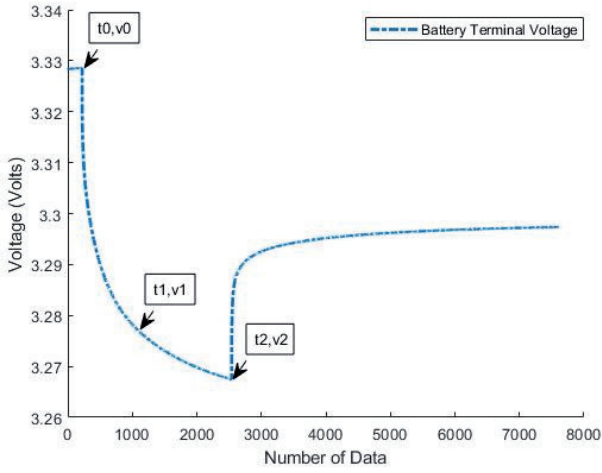


Fig. 7 Battery terminal voltage for one discharge pulse

It is easy to see that the short-term time constant effect can be observed between t_0 and t_1 , and the long-term effect is in between t_1 and t_2 . The voltage difference between v_1 and v_2 provides the value of R_l . Now, C_l can be determined as in (32), detailed in [5]:

$$C_l = \frac{\tau_l}{R_l} \quad (32)$$

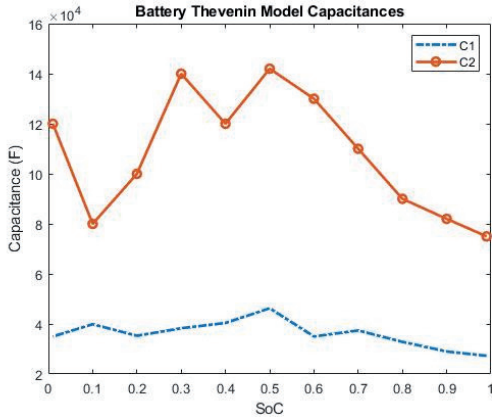


Fig. 8 C_1 and C_2 values

The capacity values in RC branches do not change significantly. The change in these values mainly depends on the ambient temperature of the battery. The study was carried out at constant room temperature and the values obtained are shown in Fig 8.

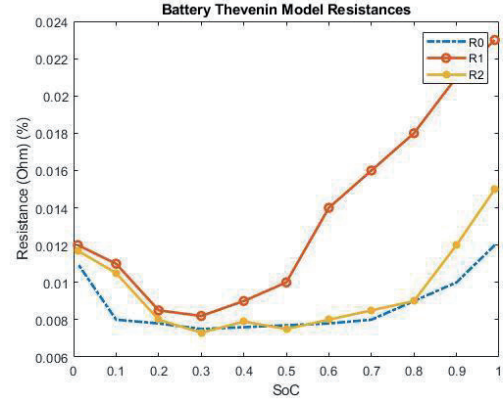


Fig. 9 R_0 , R_1 and R_2 values

4. SOC Estimation

The State of charge estimation of lithium cells can be determined by utilizing several methods and algorithms.

Determining the battery SoC is always important in a BMS as described in the Introduction part. True SoC is required to estimate the battery evaluation factor reliably and it is also optimal for energy management.

One of the method to determine SoC is the so-called Look-up table based method which uses open circuit voltage of lithium cell. This method can be used to calibrate the erroneous SoC. However, real time observation of state of charge of the lithium cell is not possible with this method since battery should be rested for long time to use this method.

Amperes-hour integral method is another way to estimate SoC of the battery. This method can provide accurate values of SoC by integrating amperes passing through battery cell according to this formula:

$$s_{0c}(k+1) = s_{0c}(k) + \int_t \eta I_l(t) \frac{1}{Q} dt \quad (33)$$

η signifies the coulombic performance of the battery cell, $I_l(t)$ specifies the current flowing through the battery cell, this value can be negative or positive depending on the flow direction, Q defined as the maximum available capacity. It ought to be stressed that Q has been indicated as the nominal limit in certain investigations. Since battery capacity is influenced by the working conditions and aging status, it should not be constant in SoC calculation. Thus, maximum capacity should be used in calculations and changed over time. One of the drawback of this method is that the initial SoC should be known so that one can add/remove the amount of SoC. Another drawback is the availability of noise and temperature drift additional to sensor noise. The combination of the above factor would decrease the reliability of this method.

Another method is the Data-driven estimation method which uses the input/output relationship of the battery cell. The method needs too much data and uses the system as a black box model. Due to the internal chemical complexity of battery,

this method would be less reliable compared to other methods. Additionally, testing and simulations in this method may also become uncontrollable.

Model based estimations are commonly used methods. The modal algorithms are Kalman Filter, Luenberger observer, PI (proportional integration) observer, H^∞ observer, sliding mode observer, et al [4]. EKF (Extended Kalman Filter) algorithm is used to estimate the SoC of the battery cell in this paper.

5. EKF Based SOC Estimation

EKF is widely used for state evaluation in nonlinear systems. Battery model must be considered as a nonlinear system and EKF can be used to state prediction for this system. Linear approximations of nonlinear functions are used for state prediction and estimation in EKF. General nonlinear system can be expressed as below:

$$x_k = f(x_{k-1} + w_k) \quad (34)$$

$$y_k = h(x_k) + v_k \quad (35)$$

Where $f(\cdot)$ and $h(\cdot)$ are nonlinear functions. These nonlinear functions should be linearized around current working point using Taylor Expression:

$$A_{k-1} = \left. \frac{\partial f(x_{k-1}, u_{k-1})}{\partial x_{k-1}} \right|_{x_{k-1} = \hat{x}_{k-1}} \quad (36)$$

$$C_k = \left. \frac{\partial h(x_k, u_k)}{\partial x_k} \right|_{x_k = \hat{x}_k} \quad (37)$$

Jacobian matrices are used in the algorithm. Jacobian matrices are denoted as A_{k-1} and C_k . A_{k-1} matrix is obtained by derivating the function $f(\cdot)$ w.r.t. x_{k-1} and u_{k-1} partially. C_k matrix is obtained by derivating the function $h(\cdot)$ w.r.t. x_k and u_k partially. EKF algorithm is given below step by step;

(1) Initialization:

Instatement of the state estimate vector x , error covariance matrix P , process noise covariance matrix Q and measurement noise covariance R .

(2) Prediction:

$$\hat{x}_{k|k-1} = A x_{k-1|k-1} + B u_{k-1} \quad (38)$$

$$P_{k|k-1} = A_k P_{k-1|k-1} A_k^T + Q \quad (39)$$

(3) Correction:

$$K = P_{k|k-1} C_k^T (C_k P_{k|k-1} C_k^T + R)^{-1} \quad (40)$$

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K (y_k - (h(x_k) + v_k)) \quad (41)$$

$$P_k = (I - K C_k) P_{k|k-1} \quad (42)$$

I is the identity matrix. Step by step EKF process runs iteratively as estimated values converge to their real values [6].

MATLAB simulations are used to calculate the parameters and solve above equations. Plots and results are obtained again in the MATLAB simulation environment.

Determining R and Q values of the algorithm is not straightforward. Some sources state that these values can be determined by parameter tuning; some sources explain this calculation based on an analytical background [7]. Process covariance matrix Q of the algorithm can be obtained as:

Recall (7);

$$B_d = \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} \quad (43)$$

$$Q = E \left(\begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} \begin{bmatrix} x & y & z \end{bmatrix} \right) = \begin{bmatrix} \alpha\alpha & \alpha\gamma & \alpha z \\ \beta x & \beta\gamma & \beta z \\ \gamma x & \gamma y & \gamma z \end{bmatrix} \quad (44)$$

From (44), x , y and z parameters are the variances of SoC, U_s and U_l respectively. Process noise covariance matrix can be calculated through this way.

R , measurement covariance matrix, can be easily determined from datasheet of the voltage measurement sensor.

These covariance matrixes effect the stability of Extended Kalman Filter directly.

We observe that increasing R results in the Kalman filter gain K to decrease. If Kalman filter gain K decreases, state of charge update will be weak. If Q increases, fluctuations increase in the SoC estimation trace [8].

6. Simulation Results

A 3.2 V Li-iron battery cell is discharged from 100% state of charge to 0% state of charge with specific periods. Current is drawn from battery cell as pulses that decrease 10% state of charge of the battery cell. After applying current pulses, battery cell is rested for approximately 2.5 hours. Fig. 10 shows SoC estimation results which are obtained by ampere-integral method and the EKF algorithm. For short usage, ampere-integral method works well and SoC value calculated by this method can be regarded as the true value.

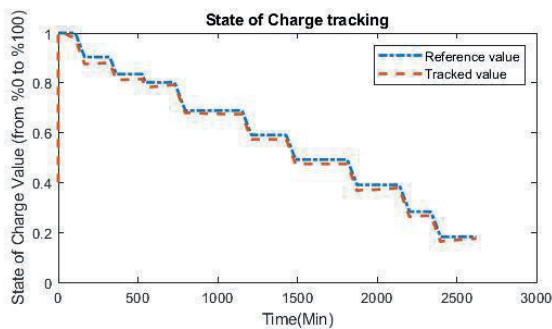


Fig. 10 State of Charge estimation w.r.t Ah-counting method

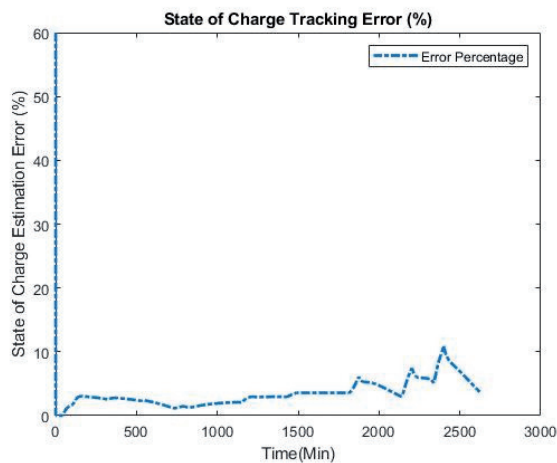


Fig. 11 State of Charge estimation error

7. Conclusion

In this paper, parameter identification is implemented by using FFRLS algorithm and time-domain method is applied for the sake of operational simplicity. SoC estimation method

based on EKF has been presented during battery discharging under constant current pulses. It can be seen clearly in the Fig. 10 that the EKF based SoC estimation curve is very close to the actual SoC curve. The maximal error between the two is lower than 6.5% in a large part of the discharge time as depicted in Fig. 11.

8. Acknowledgement

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